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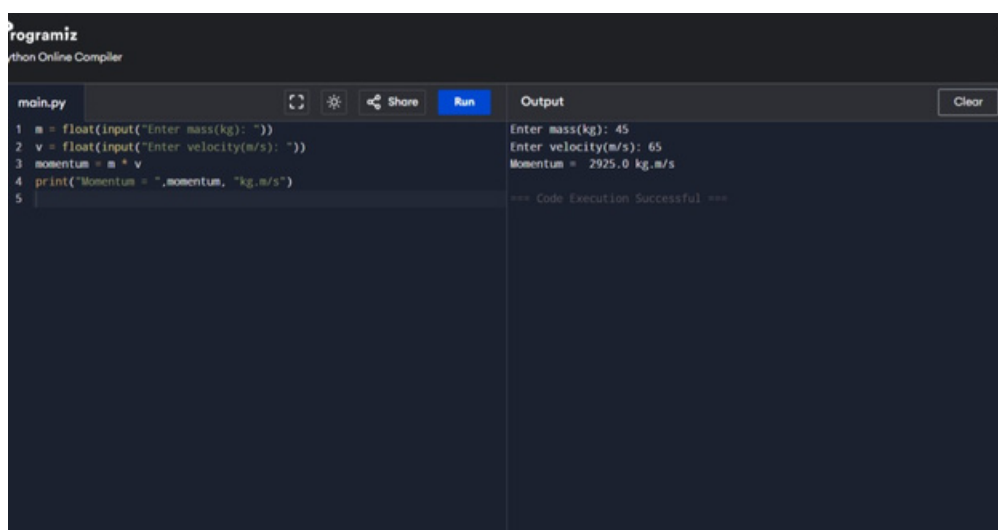
PRN : 202501080011

Branch : IT1(I11)

## **Practice Lab Assignment : 1**

1. To accept an object mass in kilograms and velocity in meters per second and display its momentum. Momentum is calculated as  $\text{momentum} = \text{mass} \times \text{velocity}$  where  $m$  is the mass of the object and  $v$  is its velocity.

Code :



The screenshot shows a web-based Python IDE. The code in the editor is as follows:

```
main.py
1 m = float(input("Enter mass(kg): "))
2 v = float(input("Enter velocity(m/s): "))
3 momentum = m * v
4 print("Momentum = ", momentum, "kg.m/s")
5
```

The output window on the right shows the following text:

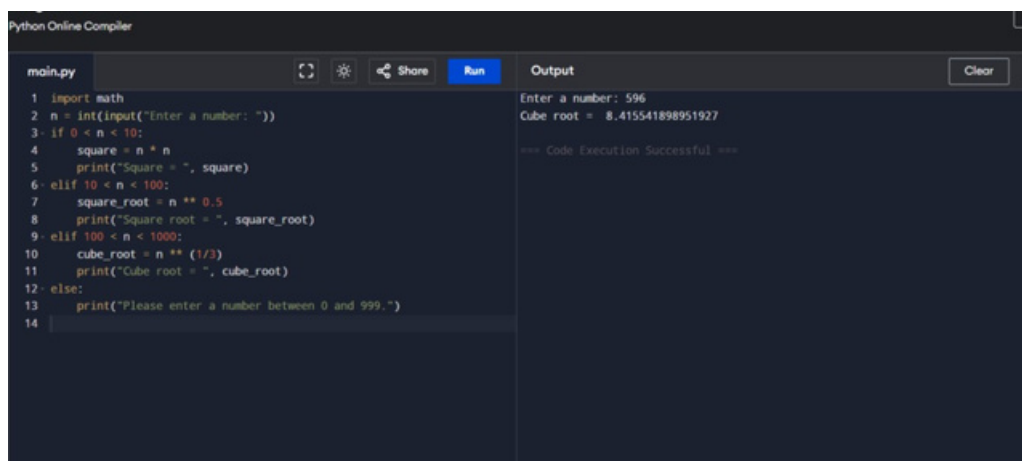
```
Enter mass(kg): 45
Enter velocity(m/s): 65
Momentum = 2925.0 kg.m/s

=== Code Execution Successful ===
```

2. Write a Python program for following conditions.

- If  $n$  is single digit print square of it.
- If  $n$  is two digit print square root of it.
- If  $n$  is three digit print cube root of it.

Code :



The screenshot shows a web-based Python IDE. The code in the editor is as follows:

```
main.py
1 import math
2 n = int(input("Enter a number: "))
3 if 0 < n < 10:
4     square = n * n
5     print("Square = ", square)
6 elif 10 < n < 100:
7     square_root = n ** 0.5
8     print("Square root = ", square_root)
9 elif 100 < n < 1000:
10    cube_root = n ** (1/3)
11    print("Cube root = ", cube_root)
12 else:
13    print("Please enter a number between 0 and 999.")
14
```

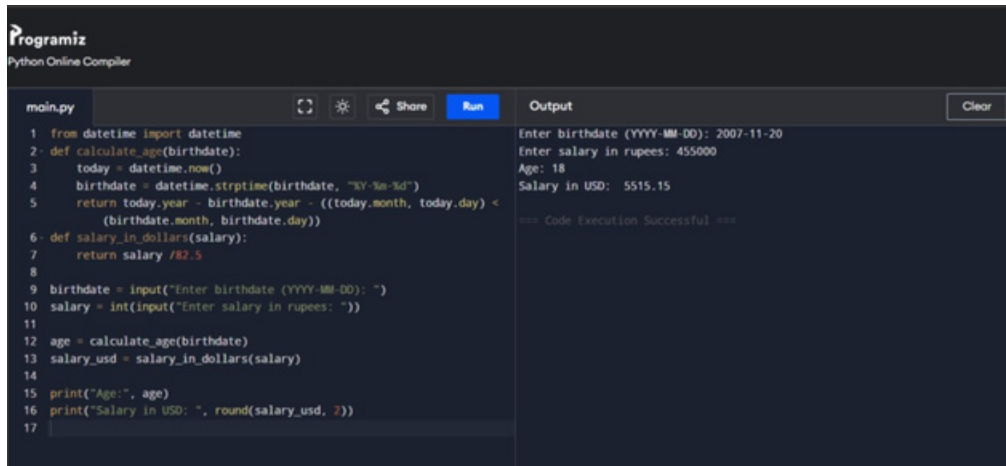
The output window on the right shows the following text:

```
Enter a number: 596
Cube root = 8.415541898951927

=== Code Execution Successful ===
```

3. Read the birthdate and salary in rupees of employees. Perform data transformation for birthdate to age and also salary which is in rupees to salary in dollars using functions.

Code :



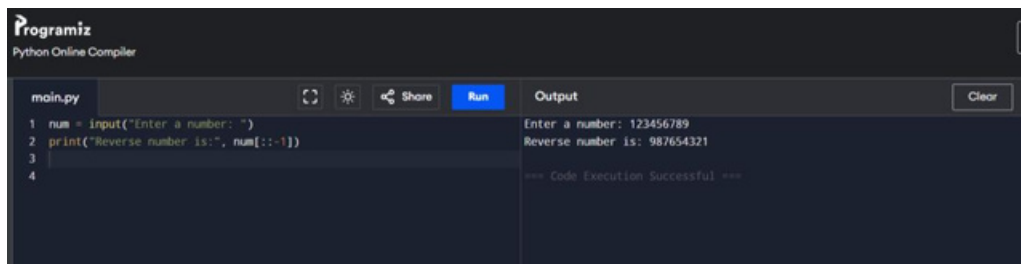
```
main.py
1 from datetime import datetime
2 def calculate_age(birthdate):
3     today = datetime.now()
4     birthdate = datetime.strptime(birthdate, "%Y-%m-%d")
5     return today.year - birthdate.year - ((today.month, today.day) <
6         (birthdate.month, birthdate.day))
7 def salary_in_dollars(salary):
8     return salary / 82.5
9 birthdate = input("Enter birthdate (YYYY-MM-DD): ")
10 salary = int(input("Enter salary in rupees: "))
11
12 age = calculate_age(birthdate)
13 salary_usd = salary_in_dollars(salary)
14
15 print("Age:", age)
16 print("Salary in USD: ", round(salary_usd, 2))
17
```

Output

```
Enter birthdate (YYYY-MM-DD): 2007-11-20
Enter salary in rupees: 455000
Age: 18
Salary in USD: 5515.15
=== Code Execution Successful ===
```

4. Print the reverse number of a given number.

Code :



```
main.py
1 num = input("Enter a number: ")
2 print("Reverse number is:", num[::-1])
3
4
```

Output

```
Enter a number: 123456789
Reverse number is: 987654321
=== Code Execution Successful ===
```

## **Lab Assignment: 1**

1. To accept students five courses marks and compute his/her result. Student is passing if he/she scores marks equal to and above 40 in each course. If student scores aggregate greater than 75 percentage, then the grade is distinction.

- If aggregate is greater than or equal to 60 and less than 75 then the grade is first division.
- If aggregate is greater than or equal 50 and less than 60, then the grade is second division.
- If aggregate is greater than or equal 40 and less than 50, then the grade is third division.

Code :

```
Programiz
Python Online Compiler

main.py
1 marks = []
2
3 for i in range(1, 6):
4     m = int(input("Enter marks of subject " + str(i) + ": "))
5     marks.append(m)
6 result = "Pass"
7 for m in marks:
8     if m < 40:
9         result = "Fail"
10        break
11 total = sum(marks)
12 percentage = total / 5
13
14 print("Total Marks:", total)
15 print("Percentage:", percentage)
16
17 if result == "Fail":
18     print("Result: Fail")
19 else:
20     print("Result: Pass")
21     if percentage > 75:
22         print("Grade: Distinction")
23     elif percentage >= 60:
24         print("Grade: First Division")
25     elif percentage >= 50:
26         print("Grade: Second Division")
27     elif percentage >= 40:
28         print("Grade: Third Division")
29
```

Output

```
Enter marks of subject 1: 65
Enter marks of subject 2: 56
Enter marks of subject 3: 56
Enter marks of subject 4: 78
Enter marks of subject 5: 87
Total Marks: 342
Percentage: 68.4
Result: Pass
Grade: First Division

=== Code Execution Successful ===
```

2. Solve the Fibonacci sequence using recursive function in Python.

Code :

```
Programiz
Python Online Compiler

main.py
1 def fib(n):
2     if n <= 1:
3         return n
4     return fib(n-1) + fib(n-2)
5
6 n = int(input("Enter number: "))
7 print(fib(n))
```

Output

```
Enter number: 6
8

=== Code Execution Successful ===
```

3. Write a Python program to print different patterns.

Code:

```
Programiz
Python Online Compiler

main.py
1 n = int(input("Enter number: "))
2 for i in range(1, n+1):
3     print("*** * i)
4
5
6
7
8
```

Output

```
Enter number: 5
*
**
***
****
*****

=== Code Execution Successful ===
```

```
Python Online Compiler

main.py
1 n = int(input("Enter number: "))
2 for i in range(n, 0, -1):
3     print("*** * i)
4
5
6
7
8
```

Output

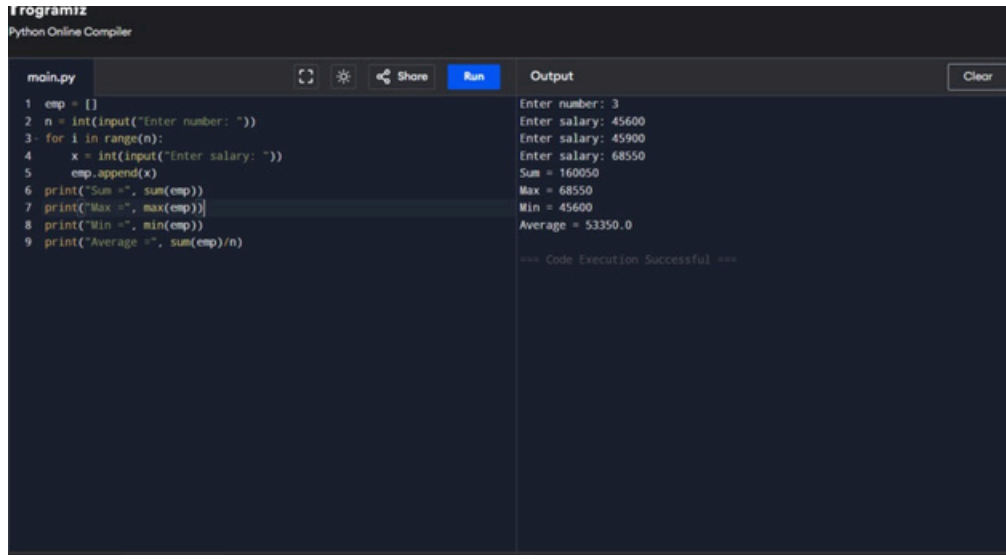
```
Enter number: 6
*****
*****
****
***
**
*

=== Code Execution Successful ===
```

# Self Study Assignment: 1

Perform all statistical analysis(Average,Max,Min,Sum)onemployee data.

Code :



The screenshot shows a web-based Python compiler interface. On the left, a code editor displays a Python script named 'main.py'. The script prompts the user to enter a number 'n', then loops 'n' times to prompt for employee salaries, storing them in a list 'emp'. It then calculates and prints the sum, maximum, minimum, and average of the salaries. On the right, the 'Output' pane shows the execution results for 'n=3' with salaries 45600, 45900, and 68550. The calculated values are Sum = 160050, Max = 68550, Min = 45600, and Average = 53350.0. A success message '=== Code Execution Successful ===' is displayed at the bottom of the output pane.

```
main.py
Python Online Compiler

1 emp = []
2 n = int(input("Enter number: "))
3 for i in range(n):
4     x = int(input("Enter salary: "))
5     emp.append(x)
6 print("Sum =", sum(emp))
7 print("Max =", max(emp))
8 print("Min =", min(emp))
9 print("Average =", sum(emp)/n)

Enter number: 3
Enter salary: 45600
Enter salary: 45900
Enter salary: 68550
Sum = 160050
Max = 68550
Min = 45600
Average = 53350.0

=== Code Execution Successful ===
```